

Autonomous and Incentivized Wireless Connection for Robust Mobile Blockchain Network

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Abstract—Blockchain has been widely implemented as a trusted platform. Previous works mainly focus on the computing capacity of devices while communication factors play a vital role in blockchain performance during dynamic wireless environments. High-speed movement causes frequent wireless connection interruptions and leads to severe performance degradation of blockchain. Besides, resource-constrained mobile devices are unwilling to selflessly contribute their energy and bandwidth for blockchain, hindering applications in dynamic mobile networks. This paper proposes a reverse auction mechanism to incentivize mobile devices to provide robust wireless connections. Devices submit their connection provision and expected rewards as bids. The mobile blockchain system uses smart contracts to autonomously execute the reverse auction to determine winners and allocate payments based on actual connections. We prove that the reverse auction mechanism is Individual Rationality (IR), Incentive Compatibility (IC), and Computational Efficiency (CE), and derive the approximation ratio 2σ of the mechanism. Extensive simulation results demonstrate that the proposed mechanism decreases up to half the energy and bandwidth consumption, but achieves a similar TPS and stale rate compared to the selfless scheme, where devices contribute all wireless connections for nothing in return. The proposed auction mechanism achieves more than 96% of the optimal social welfare.

Index Terms—Mobile blockchain, reverse auction, incentive mechanism, dynamic mobile network.

I. INTRODUCTION

WITH the maturity of 5G mobile communication technology, mobile devices (e.g., Uncrewed Aerial Vehicles (UAVs) and Uncrewed Ground Vehicles (UGVs)) have shown a monumental growth trend. Intelligent mobile devices will be the core of 6G networks [1], [2]. However, ubiquitous devices and connections led to challenges for securing data and services [3]. In 6G communication, trust requirements and implementations must be fully considered at all network and business lifecycle

stages [4]. Mobile blockchain could establish a trusted alliance of multi-party, and achieve resource sharing and independent transactions, which will be the basis for a traceable mechanism to ensure 6G trust. Integrating blockchain and mobile networks (e.g., autonomous vehicle networks, Internet of Vehicles (IoV)) not only guarantees credible data exchange but also extends the application of blockchain [5]. Besides, the decentralized characteristic of blockchains is compatible with the device-centric 6G mobile networks to construct the future Internet-of-Value. Nevertheless, applying existing blockchain architecture to wireless networks in high-speed movement is challenging.

Unlike traditional wireless networks, devices in mobile vehicle networks are characterized by high-speed movement, which means devices' wireless connections and environments change dramatically over time [6]. High-speed movement limits the communication capacity of mobile devices, and the wireless connections of devices are intermittent. It is a luxury for wireless devices to maintain robust wireless connections [7]. Classical blockchain applications (e.g., Bitcoin [8] and Ethereum [9]) are mostly designed based on a stable wired communication environment, which provides stable connections. Thus, previous works mainly focus on the effect of hash rate on blockchain performance and energy consumption [10], [11]. In contrast, the dynamic wireless environment makes communication unstable. It is nonnegligible that communication factors play a vital role in blockchain performance, sometimes even more important than the computing capacity. Specifically, high-speed movement decreases the signal strength and affects the quality and stability of block transmission. Besides, the mobility of devices results in changes in the network topology and even causes network partitioning, which influences the load balancing and performance of the blockchain network. In a nutshell, it is a prerequisite to design a mechanism to resolve the impact of high mobility on wireless connections for integrating blockchain and dynamic mobile networks.

Due to the decentralized characteristic of blockchain, an incentive mechanism is indispensable to enable devices to participate in consensus honestly. In Proof-of-Work (PoW) consensus, devices contribute their computing power to construct blocks and obtain tokens in return. Analogously, participants validate and propose blocks based on their tokens in Proof-of-Stake (PoS) and Delegated-Proof-of-Work (DPoS). These consensus mechanisms consider the consumption of devices during the consensus progress and pay corresponding rewards. However, most existing incentive mechanisms neglect the importance of block propagation. Devices that do not participate in the

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consensus consume energy and resources to help disseminate blocks without rewards, which decreases devices' enthusiasm to construct the blockchain network. In mobile networks, devices treasure energy and are unwilling to consume energy to maintain wireless connection gratis. As a result, unstable wireless connections increase the transmission delay and information loss rate, leading to a serious stale rate of blockchain. In addition, existing incentive mechanisms do not consider the impact of high mobility of devices on blockchain performance, which is the crucial factor restricting the integration of blockchain and dynamic mobile networks.

To better deploy blockchain technology in dynamic mobile networks, we investigate the mechanism to compensate mobile devices for maintaining wireless connections to propagate blocks during high-speed movement. Nevertheless, designing a mechanism adapting to dynamic mobile networks is challenging. More wireless connections bring lower latency and stale rate. However, mobile devices can not maintain too many connections restricted by bandwidth and energy resources. The block propagation latency has an upper limit. Increasing network connectivity may not consistently reduce latency, instead consumes vast bandwidth and energy. In addition, devices can not ensure that they hold certain wireless connections during high-speed movement since wireless connection disruptions are common. The heterogeneity of the dynamic mobile network also brings difficulty in designing the incentive mechanism. Heterogeneous devices have different energy costs and reward demands. Traditional unified incentive mechanisms do not consider different resource consumption and reward demands of devices and can not attract enough mobile devices to construct the blockchain. To address the above challenges, the mechanism is supposed to meet the following requirements:

- 1) Mobile blockchain should determine the appropriate network connectivity, considering both blockchain performance and network overhead.
- 2) Estimating the wireless connection change of devices during the movement, and autonomously selecting the participants to construct the mobile blockchain based on connective stability.
- 3) The incentive mechanism meets the heterogeneous requirements of devices and allocates rewards according to the actual wireless connection amounts.

In this paper, the mobile blockchain requiring robust wireless connections for stable information transmission is seen as the buyer. Mobile devices providing connections to obtain rewards are sellers. It is consistent with the typical reverse auction, which is with one buyer and many potential sellers [12]. The reverse auction is appropriate for mobile blockchains to select devices with robust wireless connections. Thus, we design a reverse auction mechanism to incentivize mobile devices to contribute to the mobile blockchain system. Devices submit their connection provisions and expected rewards as bids, and the mobile blockchain system uses smart contracts to autonomously select winners based on devices' bids and historical behaviors. The main contribution of the paper is as follows:

- We formulate the dynamic device selection problem for the mobile blockchain as an auction problem and design the

TABLE I
MAIN NOTATIONS

Notation	Definition
$\mathcal{N}$	Mobile device set
$\tilde{C}$	Expected network connectivity
Γ	Marginal welfare
$\Delta\Gamma$	Marginal security yield
ΔV	Security yield difference
h	Wireless channel gain
α	Channel fading factor
p	Transmission power
ε	Energy cost coefficient
r	Reward of bid
s	Stale rate of blockchain
γ	Utility coefficient
ω	Indicator of reverse auction winner
π	Payment of reverse auction
q	Quality of connectivity
η	Welfare loss indicator
σ	Maximal connections of bids
ϑ	Minimal quality of connectivity
$\mathbf{w}$	Winner vector

reverse auction to incentivize devices to contribute their wireless connections. In the reverse auction, we consider the impact of the device's movement on the connection interruption and propose quality of connectivity to determine winners.

- We prove that the reverse auction is Individual Rationality (IR), Incentive Compatibility (IC), and Computational Efficiency (CE). Besides, we analyze the optimality of the proposed reverse auction mechanism and derive the approximation ratio 2σ .
- Simulation results demonstrate that the reverse auction effectively maintains the performance and security of mobile blockchain systems, and the energy efficiency is at least 2 times higher than the existing schemes. Moreover, the proposed mechanism achieves more than 96% of the optimal social welfare.

The rest of this paper is organized as follows. Section II reviews the related work of the dynamic mobile network and blockchain. Section III demonstrates the mobile blockchain model in dynamic mobile networks. The proposed reverse auction mechanism is discussed in Section IV. In Section V, we give the theoretical analysis of the reverse auction mechanism. Simulation results of the performance of the designed mechanism in mobile blockchain are presented in Section VI. Finally, Section VII concludes the paper. Table I lists the main notations of this paper.

II. RELATED WORK

Recently, more work has pay attention to applying blockchain in the dynamic mobile environment. Authors in [13] proposed a blockchain radio access network for a 6G network, which provided a promising solution to enable safe and effective wireless communication. In [14], the authors developed a lightweight vehicular blockchain-enabled secure data-sharing framework in UAV-aided IoV for disaster rescue. Authors in [15] used blockchain to achieve trusted data sharing in the Internet of

Things (IoT) network. In [16], the authors propose a data aggregation method assisted by blockchain for privacy-preserving. A blockchain-envisioned approach to provide secure network connectivity for UAV communication was proposed in [17]. By using blockchain technology, the authors in [18] proposed a reliable resource allocation scheme for vehicular edge computing. The authors explored integrating blockchain with digital twin technology and applying it to digital twin scenarios, thereby enhancing the system's resource utilization efficiency and security [19] [20]. In the mobile crowdsensing system, the authors used blockchain to replace the traditional triangular architecture, ensuring data privacy and trustworthiness [21]. In [22] [23], the authors designed auction mechanisms to enhance resource utilization in edge computing, which improves the integration of edge computing with blockchain. The authors in [24] proposed a blockchain-based data-sharing framework for IoV, ensuring node privacy and data security through anonymity and auditing methods. Considering the mobility of vehicles, they design a blockchain sharding scheme to reduce communication overhead. In [25], the authors investigated the integration of blockchain and federated learning, further expanding the application scope of blockchain technology.

However, the wireless communication factors of dynamic mobile networks affect blockchain performance, which restricts the effects of integrating blockchain and dynamic mobile networks. Authors in [26] analyzed the impact of node connectivity on blockchain security, which indicated that the communication capacity also has a significant influence on the stale rate in PoW blockchain. In [27], the authors investigated block delivery in the Bitcoin network. The authors modeled the block delivery as a non-homogeneous Poisson process and derived the distribution of the block arrival rate. Authors in [28] proposed a node selection model to accelerate block transmission in the wireless IoT network. In [29], the authors analyzed the delay of the blockchain system to deduce the stale rate. The authors also considered the energy consumption of miners and found that multi-miners could reduce the energy consumption compared to a single miner. Authors in [30] investigated the minimum transmission power to execute the Practical Byzantine Fault Tolerance (PBFT) consensus protocol and analyzed the impact of node location on the blockchain performance. Besides, high mobility is a prominent characteristic of the dynamic mobile network. Previous works have proved that the mobility of nodes affects connectivity and communication performance [31], [32]. Some research focuses on the impact of high mobility on blockchain performance. In [33], the authors investigated block propagation in the IoV network and gave a closed-form expression of propagation delay. Through simulation analysis, the authors found that high mobility could speed up the block propagation. Authors in [34] concluded that the throughput of blockchain is significantly affected by the node movement and the transmission power be beyond a certain threshold to ensure the block delivery. In [35], the authors investigated the impact of mobility on blockchain performance in IoV networks. The authors indicated that nodes have little time to exchange block information during the movement, which causes a high stale rate and decreases the throughput in the blockchain system.

Based on previous works, dynamic mobile communication environments challenge the integration of blockchain and mobile networks. In this paper, we mainly focus on designing the incentive mechanism in the mobile blockchain system to suit dynamic mobile networks. Incentive mechanisms can guide the behavior of devices to improve the blockchain system performance [36]. In Ethereum, the Casper protocol is proposed to accord with the consensus transformation [37]. The incentive mechanism in Casper provides safety and liveness for the Ethereum system by using both reward and penalty mechanisms [38]. Furthermore, the Gasper protocol is designed to accommodate the PoS consensus in Ethereum [39]. Some researchers have utilized the incentive mechanism to assist blockchain applications in wireless networks. In [40], authors used contract theory to provide different rewards for heterogeneous IoT devices. In the proposed contract, devices that consume more energy will obtain higher rewards. Authors in [41] improved the deposit mechanism of Ethereum, which is more appropriate for IoT devices. Auction is a classical theory for designing the incentive mechanism and is widely used in mobile scenarios [42]. In [43], the authors proposed an auction mechanism to allocate computing resources in PoW blockchain networks. Authors in [44] designed a dual auction mechanism to encourage light nodes to forward transactions for the blockchain system. Although there are already works on using incentive mechanisms to improve the performance of mobile blockchain networks, few studies have investigated the impact of high mobility on blockchain performance and redesigned the corresponding mechanism to resolve this.

III. SYSTEM MODEL

In this section, we demonstrate our mobile blockchain model in dynamic UAV networks. As shown in Fig. 1, mobile devices (i.e., UAVs) sense environment data for monitoring ecology. Devices use mobile blockchain as a trusted platform to record and exchange data. However, the wireless connections are unstable due to the device movement, which impedes the propagation of blockchain information and impairs the blockchain performance. We propose the reverse auction to address the problem. Mobile devices submit their connection provision and expected rewards as bids to the smart contract, which determines the winners based on bids and quality of connectivity autonomously. Then, mobile devices have more impetus to contribute transmission power to build robust wireless connections. Once the new block, the smart contract updates the quality of connectivity and allocates rewards for devices.

A. Mobile Blockchain System in Dynamic Mobile Networks

We consider the blockchain scenario with various heterogeneous mobile devices. Each device connects to others through wireless channels during high-speed movement. Since the mobility of devices, the wireless connections among devices are unstable and the network topology varies dynamically with time. The mobile blockchain system adopts the PoW consensus mechanism, which is the most classical blockchain architecture. We define the device set as $\mathcal{N}$ and assume that there are $N = |\mathcal{N}|$

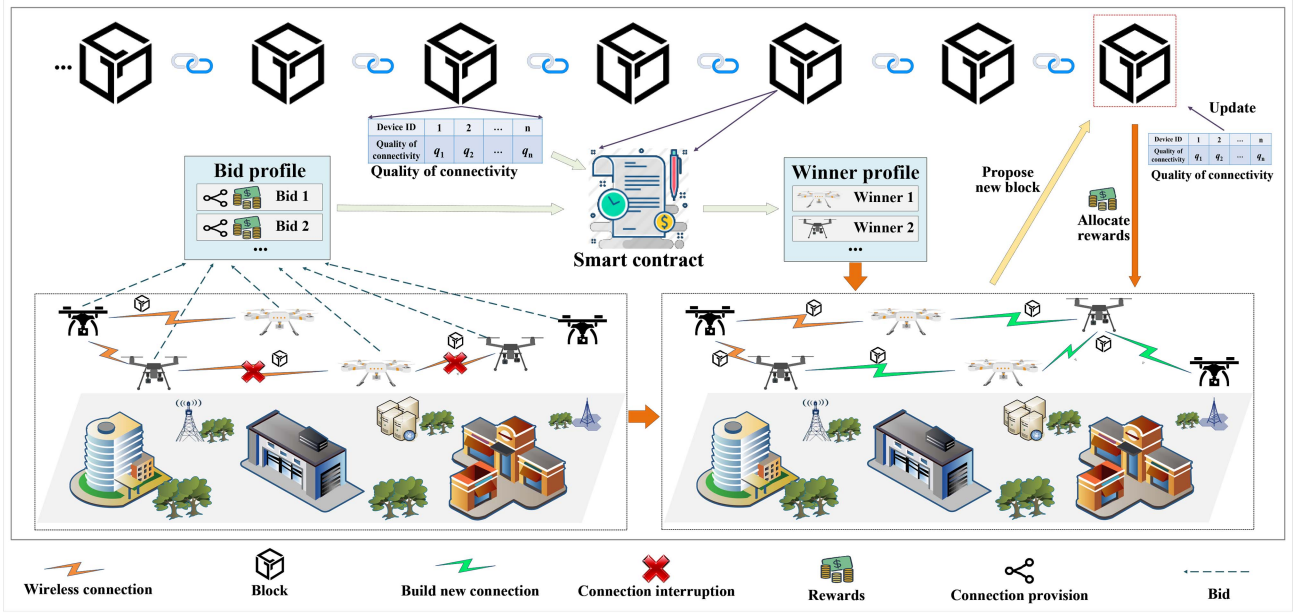


Fig. 1. The reverse auction for mobile blockchain system in the dynamic mobile network.

devices in the dynamic mobile network. All devices are selfish and rational, and they choose whether to join the blockchain construction based on revenue. Traditional blockchain systems focus on incentivizing devices to operate the consensus mechanism and validate blocks. We mainly investigate the incentive mechanism to encourage mobile devices to maintain robust wireless connections to propagate blocks.

In the proposed reverse auction, the mobile devices act as sellers (bidders), which are capable of providing wireless connections to earn rewards. The developer of mobile blockchain acts as the buyer in the auction, aiming to incentivize mobile devices to share their connections through reward payments, thereby maintaining robust network connectivity and ensuring efficient information propagation. The smart contract, serving as the auctioneer, collects bids from mobile devices and determines the winning devices and corresponding rewards based on the auction algorithm. The mobile device i submits the bid (c_i, r_i) , where c_i is the wireless connections established by device i , and r_i is the expected reward. The mobile blockchain system selects devices to satisfy the network connectivity demand and allocates rewards. Note that our proposed reverse auction mechanism works as a complementary layer to existing incentive mechanisms rather than replacing them. Existing Pow-based mechanisms reward devices for their consensus efforts. The proposed mechanism enhances the overall incentive structure from the perspective of network connectivity and is compatible with existing mechanisms.

The device moves continuously according to its tasks. The movement of devices affects their distance, which brings changes in the Signal-to-Noise Ratio (SNR). Considering the effects of fading in wireless channels, we use α to denote the channel fading factor. According to the standard free space path loss model [45], we define the SNR of device i to device j in

the following:

$$h_{ij} = A_d \left(\frac{\lambda}{4\pi d_{ij}} \right)^{d_e}, \quad (1)$$

$$SNR_{ij} = \frac{S_{ij}}{n_0} = \frac{\alpha_{ij} h_{ij} p_{ij}}{n_0}, \quad (2)$$

where $\lambda = \frac{3 \times 10^8}{f_c}$ is the wavelength, h_{ij} is the wireless channel gain between devices i and j , A_d accounts for the combined antenna gains, d_{ij} is the distance between devices i and j , f_c denotes the carrier frequency, d_e denotes the path loss exponent, p_{ij} is the transmission power of device i to j , and n_0 denotes the noise power. We set τ as the SNR threshold. Only devices with a SNR higher than the threshold τ can communicate with each other.

B. Analysis of Wireless Connectivity Impact on Mobile Blockchain Performance

In the mobile blockchain network, devices utilize Peer-to-Peer (P2P) communication methods (the gossip protocol) for block transmission. When a device successfully mines a block, it needs to broadcast this block to other devices for validation and chain extension. Consensus is achieved after all devices receive and confirm the block, enabling the mining of the block at the next height.

When devices propose new blocks, security is primarily measured by the stale rate. The stale rate is closely related to forking events, which occur when multiple blocks at the same height propagate simultaneously through the network. Since the main chain can only contain one block at each height, the remaining blocks become stale. We define the stale rate s as:

$$s = \frac{B_s}{B}, \quad (3)$$

where B_s is the number of discarded blocks, and B is the total number of blocks in the network. The block propagation relies heavily on network connectivity. When wireless connections become unstable due to device mobility, it can cause significant delays in block propagation and even lead to temporary network partitioning. This directly affects the mobile blockchain's confirmation latency and potentially increases the stale rate.

Then, we derive the formula between the network connectivity and forking events. According to [26], the forking probability F is defined as:

$$F = 1 - (1 - h) \int_0^\infty \bar{u}_r(t) dt, \quad (4)$$

where h is the merged Poisson process of rate representing the block generation of whole network, and $\bar{u}_r(t)$ is the average block propagation delay. The average block propagation delay is related to the network connectivity [46]. According to the reserach [47], the average block propagation delay could be modeled as:

$$\bar{u}_r(t) = 1 - \exp\left(-\frac{1}{\varrho}\right), \quad (5)$$

where ϱ is the network strength parameter related to the number of network connections, the bandwidth efficiency, and the communication quality among devices.

In the dynamic wireless network, device mobility is the most prominent characteristic. The wireless connections become unstable due to device mobility, which causes significant delays in block propagation. This directly affects the mobile blockchain's consensus latency and potentially increases the forking events and stale rate. For example, if a mined block cannot be propagated quickly enough due to poor wireless connectivity, another device might mine a competing block, resulting in high stale rate. Therefore, maintaining robust connectivity throughout the network is crucial for the performance and security of the mobile blockchain system.

C. Mobile Device Model

As mentioned above, each mobile device is in a state of high-speed movement. We use the coordinate (x_i^t, y_i^t) to denote the location of device i at the time frame t . The change of position is follows the following equation:

$$\begin{aligned} x_i^{t+1} &= x_i^t + v_i \Delta t, \\ y_i^{t+1} &= y_i^t + v_i \Delta t, \end{aligned} \quad (6)$$

where v_i is the velocity of device i , and Δt is the unit time interval. The distance between devices i and j at time t is calculated:

$$d_{ij}^t = \sqrt{(x_i^t - x_j^t)^2 + (y_i^t - y_j^t)^2}. \quad (7)$$

Mobile devices maintain robust connections by increasing the transmission power to decrease the block propagation delay, which consumes additional energy. Mobile blockchain distribute rewards for devices in return. Besides, devices have heterogeneous energy costs. We define the utility function of the mobile

device i as follows:

$$u_i = r_i - \varepsilon_i p_i^2, \quad (8)$$

where ε_i is the energy cost coefficient, and p_i is the transmission power. Each device evaluates the value of utility before participating in the mobile blockchain. Only devices with non-negative utility will participate in blockchain construction.

D. Mobile Blockchain Model

From the perspective of the mobile blockchain, more network connections bring lower delay and stale rate, which enhances the security of the blockchain system. Determining the impact of network connectivity on the stale rate is the prerequisite to estimating the mobile blockchain utility. We propose the function to characterize the relationship between network connectivity and stale rate in the following:

$$s = \frac{\beta_1}{C + \beta_2} + \beta_3, \quad (9)$$

where s is the stale rate, $C = \sum_{i=1}^N c_i$ is the total connections of the blockchain network, β_1 , β_2 , and β_3 are coefficients. Curve fitting is a widely adopted approach in wireless network modeling [48]. We conduct extensive experiments to collect empirical data on the relationship between network connectivity and stale rate. Then, we use curve fitting to derive coefficients β_1 , β_2 , and β_3 .

In the PoW-based blockchain, the network utility includes security and performance [49]. Stale rate, as a comprehensive performance indicator, directly reflects the blockchain security and correlates with TPS and resource efficiency. Thus, we select stale rate as the main utility metric for mobile blockchain. The mobile blockchain incentivizes devices to consume energy to hold stable connections by allocating corresponding rewards. Thus, the utility function of mobile blockchain is defined as:

$$u_B = \gamma(1 - s) - \sum_{i=1}^N \omega_i r_i. \quad (10)$$

where γ is the utility coefficient, ω_i is the indicator to denote whether device i is selected. The first term in (10) denotes the security yield brought by the non-stale rate, and the second term is the reward distributed to devices.

Considering the dynamic changes in network connectivity, devices can't ensure the declared wireless connections c_i during the movement. We introduce the factor $\tilde{c}_i$ to determine the actual connection provision of device i . Accordingly, we use quality of connectivity q_i to measure the utility contributed by device i to the blockchain network and allocate rewards based on the quality of connectivity, where $\pi_i = q_i r_i$ is the payment for device i . We provide a detailed description of the determination process for the quality of connectivity q_i in Section IV. Then, we rewrite the utility function of mobile blockchain in the following:

$$u_B = \gamma \left(1 - \left(\frac{\beta_1}{\sum_{i=1}^N \omega_i \tilde{c}_i + \beta_2} + \beta_3 \right) \right) - \sum_{i=1}^N \omega_i \pi_i. \quad (11)$$

E. Problem Formulation

By introducing the quality of connectivity, we also rewrite the utility function of device i :

$$u_i = \pi_i - \varepsilon_i p_i^2. \quad (12)$$

According to (11) and (12), the social welfare of the mobile blockchain system is derived in the following:

$$\begin{aligned} U_S &= u_B + \sum_{i=1}^N \omega_i u_i \\ &= \gamma \left(1 - \left(\frac{\beta_1}{\sum_{i=1}^N \omega_i \tilde{c}_i + \beta_2} + \beta_3 \right) \right) - \sum_{i=1}^N \omega_i \varepsilon_i p_i^2. \end{aligned} \quad (13)$$

Aiming to maximize the social welfare, the problem is formalized as follows:

$$\max_{\omega} U_S, \quad (14)$$

$$s.t. \omega_i \in \{0, 1\}, \quad \forall i \in \mathcal{N}, \quad (15)$$

$$q_i \in [0, 1], \quad \forall i \in \mathcal{N}, \quad (16)$$

$$0 \leq p_i \leq p_{\max}, \quad \forall i \in \mathcal{N}, \quad (17)$$

$$\sum_{i=1}^N c_i \geq \bar{C}, \quad (18)$$

where $p_{\max}$ denotes the maximal transmission power of devices, and $\bar{C}$ denotes the demand for network connectivity, which is the tradeoff between mobile blockchain performance and resource consumption. Equation (17) is the quality of connectivity constraint, (18) represents that the transmission power of the device can not exceed the maximal power, and (18) denotes that the total wireless connections of selected mobile devices must meet the demand of the network connectivity.

It is easily verified that the social welfare maximization problem is NP-hard, and we present the proposed approximate algorithm in the next section to solve the problem.

IV. REVERSE AUCTION MECHANISM DESIGN

In this section, we demonstrate our proposed reverse auction mechanism to maximize the social welfare of the mobile blockchain system. First, we use the quality of connectivity q to measure the contribution of the device's connection provision on the security yield of mobile blockchain. Then, we design the winner device selection algorithm to select mobile devices with high quality of connectivity and lower rewards while meeting the expected network connectivity requirement of mobile blockchain. Last, we propose a pricing algorithm to determine payments based on the quality of connectivity.

In each reverse auction epoch, mobile devices submit bids to the smart contract, which uses bids and previous quality of connectivity to select winners. After finishing transmission tasks of the existing epoch, the smart contract calculates the quality of connectivity for each winning device and determines

Algorithm 1: Quality of Connectivity Determination.

Input: The expected network connectivity $\bar{C}$, bid set $\mathbf{b}$, and actual connection provision vector $\tilde{\mathbf{c}}$.

Output: The quality of connectivity vector $\mathbf{q}$.

1: **Initialization**

2: $q_i \leftarrow 1$

3: **begin**

4: $\Delta V \leftarrow \gamma \left(\frac{\beta_1}{\sum_{i=1}^N \tilde{c}_i + \beta_2} - \frac{\beta_1}{\bar{C} + \beta_2} \right)$

5: **if** $\Delta V > 0$ **then**

6: $\mathcal{O} \leftarrow \emptyset$

7: **for each** i **and** $\tilde{c}_i \neq c_i$ **do**

8: $\mu_i \leftarrow \frac{r_i}{\tilde{c}_i}$

9: $\mathcal{O} \leftarrow \mathcal{O} \cup \{i\}$

10: **end for**

11: **for** $i \in \mathcal{O}$ **do**

12: $\theta_i \leftarrow \frac{\mu_i \Delta V}{\sum_{j \in \mathcal{O}} \mu_j}$

13: $q_i \leftarrow 1 - \frac{\theta_i}{r_i}$

14: **end for**

15: **end if**

16: **end**

the final payments by the latest quality of connectivity. Mobile blockchain uses a smart contract to select devices with robust wireless connections autonomously, which ensures stable blockchain information propagation and expected state rate under dynamic networks.

A. Quality of Connectivity Determination

Considering the wireless channel uncertainty and the high mobility of devices, assessing the quality of the wireless connection provided by devices during motion is critical. The quality of connectivity q is used to evaluate the actual performance of the device's connections, serving as the basis for reward allocation and device selection of the mobile blockchain system. We expound on the determined process of quality of connectivity in Algorithm 1.

While completing a block confirmation, the mobile blockchain system examines the actual wireless connections of devices and calculates the security yield difference between the expected network connectivity $\bar{C}$ and the actual established connections $\sum_{i=1}^N \tilde{c}_i$ (line 4). The security yield difference $\Delta V > 0$ means that there existing devices fail to hold the required wireless connections during the block propagation process. To quantify the quality of connectivity q_i , the mobile blockchain system first calculates the unit connection reward μ_i and identifies the devices that do not meet the connectivity requirements (lines 5-10). The unit connection reward μ_i denotes the compensation demanded by device i for delivering the unit connection in the mobile blockchain network. Then, the mobile blockchain system obtains the security yield loss θ_i brought by device i , which is used to determine the quality of connectivity q_i (lines 11-13). The security yield loss θ represents that devices with high unit connection rewards should afford a greater proportion of the security yield loss, which incurs more reward reduction.

Algorithm 2: Winner Devices Selection.

Input: The expected network connectivity $\bar{C}$, device set $\mathcal{N}$, bid set $\mathbf{b}$, and historical quality of connectivity set $\mathbf{q}^{t-1}$.
Output: The winner devices vector $\mathbf{w}$, the non-negative marginal welfare devices vector $\mathbf{w}^+$, and the marginal security yield set $\mathcal{L}$.

- 1: **Initialization**
- 2: $\mathbf{w} \leftarrow \emptyset$
- 3: $\mathcal{F} \leftarrow \emptyset$
- 4: **begin**
- 5: **for each** $i \in \mathcal{N}$ **do**
- 6: $\tau_i \leftarrow \frac{q_i^{t-1} c_i}{r_i}$
- 7: $\mathcal{F} \leftarrow \mathcal{F} \cup \{\tau_i\}$
- 8: **end for**
- 9: Sort $\mathcal{F}$ in descending order $\{\tau_1, \tau_2, \dots, \tau_{|\mathcal{F}|}\}$
- 10: Record corresponding identity $\mathcal{I} = \{I_1, I_2, \dots, I_{|\mathcal{F}|}\}$
- 11: $i \leftarrow I_1, \hat{\mathbf{w}} \leftarrow \emptyset, \mathcal{L} \leftarrow \emptyset$
- 12: $\hat{\mathbf{w}} \leftarrow \hat{\mathbf{w}} \cup \{i\}$
- 13: $\Delta \Gamma_i \leftarrow \gamma(1 - (\frac{\beta_1}{c_i + \beta_2} + \beta_3))$
- 14: $\Gamma \leftarrow q_i^{t-1} \Delta \Gamma_i - r_i, \mathcal{L} \cup \{\Delta \Gamma_i\}$
- 15: **While** $\Gamma \geq 0$ **do**
- 16: $\mathbf{w} \leftarrow \mathbf{w} \cup \{i\}, \mathcal{I} \leftarrow \mathcal{I} \setminus \{i\}$
- 17: $i \leftarrow i \oplus 1$
- 18: $\hat{\mathbf{w}} \leftarrow \hat{\mathbf{w}} \cup \{i\}$
- 19: $\Delta \Gamma_i \leftarrow \gamma \left(\frac{\beta_1}{\sum_{j=1}^{|\mathbf{w}|} c_j + \beta_2} - \frac{\beta_1}{\sum_{j=1}^{|\mathbf{w}|} c_j + c_i + \beta_2} \right)$
- 20: $\Gamma \leftarrow q_i^{t-1} \Delta \Gamma_i - r_i, \mathcal{L} \cup \{\Delta \Gamma_i\}$
- 21: **end while**
- 22: $\mathbf{w}^+ \leftarrow \mathbf{w}$
- 23: **While** $\sum_{i=1}^{|\mathbf{w}|} c_i < \bar{C}$ **do**
- 24: $c' \leftarrow \bar{C} - \sum_{i=1}^{|\mathbf{w}|} c_i$
- 25: $l = \arg \max_{i \in \mathcal{I}} \frac{\gamma \left(\frac{\beta_1}{\sum_{j=1}^{|\mathbf{w}|} c_j + \beta_2} - \frac{\beta_1}{\sum_{j=1}^{|\mathbf{w}|} c_j + c_i + \beta_2} \right) - r_i}{q_i^{t-1} \min\{c_i, c'\}}$
- 26: $\mathbf{w} \leftarrow \mathbf{w} \cup \{l\}, \mathcal{I} \leftarrow \mathcal{I} \setminus \{l\}$
- 27: **end while**
- 28: **end**

B. Winner Devices Selection

After receiving the bids of devices, the mobile blockchain system needs to select winners to construct the network. From the perspective of maximizing social welfare, the mobile blockchain system prefers devices with more connections and fewer rewards. Besides, the historical quality of connectivity is a significant factor in selecting devices. We consider both the historical quality of connectivity and devices' bids to select winners. Algorithm 2 demonstrates the winner selection mechanism of reverse auction. First, mobile blockchain sorts devices based on bids and historical quality of connectivity (lines 5-10). Then, the mobile blockchain identifies whether the marginal security yield $\Delta \Gamma$ brought by the devices is higher than the reward paid (lines 11-21). We define the marginal welfare Γ as the difference between the marginal security yield and the reward (lines 14, 20). Through calculating the ratio τ and sorting the devices, the

algorithm prioritizes selecting devices that can provide higher marginal yield. Thus, the selection order of the mobile devices ensure that a winner set with the highest marginal benefits can be obtained. After selecting all devices with non-negative marginal welfare, the mobile blockchain system judges whether the connectivity provided by devices meets the expected network connectivity. The loop (lines 23-27) aims to find the mobile device with minimum marginal welfare loss until satisfying the expected network connectivity. Note that mobile blockchain may obtain connectivity more than the expected network connectivity (lines 15-21), but these winning devices with non-negative marginal welfare, and the mobile blockchain system is willing to get them to construct the blockchain network.

C. Reverse Auction Pricing

While completing the block confirmation, the smart contract autonomously determines payments according to the actual performances of devices. Algorithm 3 demonstrates the process of reverse auction pricing. The mobile blockchain first runs Algorithm 1 to obtain the quality of connectivity of winner devices (line 4). For every device i of winner vector $\mathbf{w}$, the mobile blockchain executes Algorithm 1 with device set $\mathcal{N}$ except device i to get a new winner vector $\mathbf{w}'$ (lines 5-7). Then, the mobile blockchain system compares the number between the original non-negative marginal welfare devices vector $\hat{\mathbf{w}}$ and the new winner vector $\mathbf{w}'$. There are two cases in the following:

- **Case 1** (lines 9-10): Devices in $\mathbf{w}'$ are with non-negative marginal welfare. Mobile blockchain could satisfy the network connectivity without device i and has no marginal welfare loss, which means that device i is replaceable. Thus, the payment is the marginal security yield brought by the devices i .
- **Case 2** (lines 12-17): Some devices bring marginal welfare loss in the new winner vector $\mathbf{w}'$. In this case, mobile blockchain can't achieve the expected network connectivity by selecting all non-negative marginal welfare devices without the device i . If device i is substituted, the mobile blockchain system suffers the welfare loss. Therefore, the device i deserves the payment that is equal to replacing the device with the maximum marginal welfare loss.

D. Smart Contract Implement

The smart contracts in our system are deployed on the mobile blockchain network. Specifically, the smart contract code is stored as part of the blockchain's state, and each mobile device participating in the network maintains a copy of these contracts through blockchain synchronization. This approach is similar to Ethereum's smart contract deployment mechanism, where contracts become an integral part of the distributed ledger.

The communication between devices and the smart contract occurs through transaction-based interactions. Smart contract is initially deployed through special transactions that contain the contract bytecode. Once deployed, the contract receives a unique address on the mobile blockchain, and the contract's state and code are replicated across all participating devices. Devices interact with smart contracts by sending transactions to

Algorithm 3: Payment Pricing.

Input: The device set $\mathcal{N}$, winner devices vector $\mathbf{w}$, the non-negative marginal welfare devices vector $\mathbf{w}^+$, and the marginal security yield set $\mathcal{L}$.

Output: The payment vector π .

- 1: **Initialization**
- 2: $\pi \leftarrow \{0, \dots, 0\}$
- 3: **begin**
- 4: $\triangleright$ **Quality of Connectivity Determination**
- 5: **for each** $i \in \mathbf{w}$ **do**
- 6: $\triangleright$ **Winner Devices Selection on** $\mathcal{N} \setminus \{i\}$
- 7: $\mathbf{w}' \leftarrow$ the new winner vector
- 8: $\hat{\mathbf{w}} \leftarrow \mathbf{w}^+ \setminus \{i\}$
- 9: **if** $|\mathbf{w}'| < |\hat{\mathbf{w}}|$ **then**
- 10: $\pi_i \leftarrow q_i \gamma (1 - (\frac{\beta_1}{c_i + \beta_2} + \beta_3))$
- 11: **else**
- 12: $\omega_i \leftarrow \gamma \left(\frac{\beta_1}{\sum_{j=1}^{|\hat{\mathbf{w}}|} c_j + \beta_2} - \frac{\beta_1}{\sum_{j=1}^{|\hat{\mathbf{w}}|} c_j + c_i + \beta_2} \right)$
- 13: $c' \leftarrow \bar{C} - \sum_{i=1}^{|\hat{\mathbf{w}}|} c_i$
- 14: $\pi_i \leftarrow q_i \gamma (1 - (\frac{\beta_1}{c_i + \beta_2} + \beta_3))$
- 15: **for each** $k \in \mathbf{w}' \setminus \hat{\mathbf{w}}$ **do**
- 16: $\omega_k \leftarrow \gamma \left(\frac{\beta_1}{\sum_{j=1}^{|\hat{\mathbf{w}}|} c_j + \beta_2} - \frac{\beta_1}{\sum_{j=1}^{|\hat{\mathbf{w}}|} c_j + c_k + \beta_2} \right)$
- 17: $\pi_i \leftarrow \max \left\{ \pi_i, \omega_i + (r_k - \omega_k) \frac{q_i \min\{c_i, c'\}}{q_k \min\{c_k, c'\}} \right\}$
- 18: **end for**
- 19: **end if**
- 20: **end for**
- 21: **end**

the contract's address. These transactions can include function calls and bids. The contract execution is triggered by these transactions and processed by all devices maintaining the mobile blockchain, and the execution results (winner set and pricing scheme) are recorded in new blocks and propagated through the network.

The contract's state is updated based on the execution results, and all devices maintain a consistent view of the contract state through blockchain consensus. State changes are only valid when confirmed through the PoW consensus mechanism. This design ensures decentralized execution with no single point of failure, consistent contract state where all devices have the same view of contract operations, secure interactions verified through consensus, and autonomous operation where smart contracts execute automatically when conditions are met.

V. PERFORMANCE ANALYSIS OF PROPOSED REVERSE AUCTION MECHANISM

In this Section, we demonstrate that our proposed reverse auction mechanism is IR, IC, and CE. Moreover, we derive the approximation ratio of the reverse auction.

A. Individual Rationality

First, we prove the proposed reverse auction is IR, which means wireless devices receive non-negative utility in the auction.

Theorem 1: The reverse auction mechanism is Individual Rationality for mobile devices.

Proof: Before submitting bids for the mobile blockchain network, devices evaluate their utilities. Since all devices are rational and selfish, they determine $r \geq \varepsilon p^2$ to pursue non-negative utilities. In the winner selection phase, winning devices consume energy to provide wireless connections and get returns. Devices that fail to bid have no energy cost and obtain zero utility. Thus, we prove that winners will receive non-negative utilities.

From Algorithm 3, we observe that the payment of winning device i has two formats.

Case 1: $\pi_i = q_i \Delta \Gamma_i$. In this case, the reward of device i is $\Delta \Gamma_i$. In Algorithm 2 (lines 14-20), only the marginal security yield $\Delta \Gamma_i$ is higher than the supposed reward, the device i has the chance to win the auction. We can derive

$$\Delta \Gamma_i \geq r_i \geq \varepsilon p^2. \quad (19)$$

The payment of device i is higher than the cost and gets positive utility.

Case 2: $\pi_i = \max\{q_i \Delta \Gamma_i, \omega_i + (r_k - \omega_k) \frac{q_i \min\{c_i, c'\}}{q_k \min\{c_k, c'\}}\}$. In this case, the minimum reward of device i is the marginal security yield $\Delta \Gamma_i$, which ensures the device gets the positive utility.

Therefore, the winning devices could receive non-negative utilities through contributing connections for the mobile blockchain network, and Theorem 1 is proved. ■

B. Incentive Compatibility

Next, we give the IC proof of the proposed mechanism.

Theorem 2: The reverse auction mechanism is Incentive Compatibility for mobile devices.

Proof: To ensure the reverse auction is IC, we prove that the device can not submit a false bid to obtain extra profit, and the optimal bid strategy is honest bidding, i.e., $r_i = \varepsilon_i p_i^2$. In the bidding phase, the device i provides the expected connections c_i and corresponding rewards r_i . We prove whether the device i wins the auction or not, there is no utility to be made from submitting the false bid in the following.

1) $\Gamma_i \geq 0$: While the marginal welfare $\Gamma_i \geq 0$, the device i wins the auction. To pursue a higher utility, the device i has the chance to submit a false bid (c_i, r'_i) . If the false bid does not alter the sorted set $\mathcal{F}$, the device still brings the marginal security yield $\Delta \Gamma_i$. While $r'_i \geq \Delta \Gamma_i$, the device loses the auction and obtain 0 utility. Thus, the false rewards $r_i \leq r'_i \leq \Delta \Gamma_i$ or $r'_i \leq r_i$. Based on the payment pricing, the device i has two payment formats.

Case 1: $\pi_i = q_i \gamma (1 - (\frac{\beta_1}{c_i + \beta_2} + \beta_3))$. According to the (12), the utility function of device i is

$$u_i = q_i \gamma \left(1 - \left(\frac{\beta_1}{c_i + \beta_2} + \beta_3 \right) \right) - \varepsilon_i p_i^2, \quad (20)$$

where $\gamma(1 - (\frac{\beta_1}{c_i + \beta_2} + \beta_3))$ is the maximal marginal security yield brought by the device. The device's utility is related to its provided wireless connections and not affected by the sorting order. The device i can not get higher utility through submitting a false bid in this case.

Case 2: $\pi_i = \max\{q_i \gamma (1 - (\frac{\beta_1}{c_i + \beta_2} + \beta_3)), \omega_i + (r_k - \omega_k) \frac{q_i \min\{c_i, c'\}}{q_k \min\{c_k, c'\}}\}$. In this case, the device i is the vital device and

could obtain the payment higher than its marginal security yield. We rewrite the utility function of device i

$$u_i = \max \left\{ q_i \gamma \left(1 - \left(\frac{\beta_1}{c_i + \beta_2} + \beta_3 \right) \right), \omega_i + (r_k - \omega_k) \frac{q_i \min \{c_i, c'\}}{q_k \min \{c_k, c'\}} \right\} - \varepsilon_i p_i^2, \quad (21)$$

where indicates that the utility of device i is just related to the provided connections and the device with maximum marginal welfare loss. The proposed pricing mechanism has compensated for the extra reward. Submitting the false expected reward r'_i does not improve the utility.

If the false bid alters the sorted set $\mathcal{F}$, the device brings the marginal security yield $\Delta \Gamma'_i$. From the above proof, the device's utility is independent of the sorting order. The final payment is also $q_i \gamma (1 - (\frac{\beta_1}{c_i + \beta_2} + \beta_3))$ or $\max \{ q_i \gamma (1 - (\frac{\beta_1}{c_i + \beta_2} + \beta_3)), \omega_i + (r_k - \omega_k) \frac{q_i \min \{c_i, c'\}}{q_k \min \{c_k, c'\}} \}$. Thus, the fals bid can not obtain extra rewards.

2) $\Gamma_i < 0$: While the marginal welfare $\Gamma_i < 0$, the device i has the following cases:

Case 1: Device i loses the auction. If the device i submit a higher bid, it still loses the auction and obtains zero utility. Only the device i that decreases the expected reward may win the auction. Due to selfish and rational characteristics, the minimum expected reward of device i is equal to the energy cost, which means an honest bid is the dominant strategy.

Case 2: Device i wins the auction. Similar to the situation $\Gamma_i \geq 0$, the payment for the device is not determined by its expected reward and the sorting order. Changing the expected rewards can not bring higher utility for the device.

Based on the above analysis, the bids of devices are related to the winner selection of the auction but do not determine the payment, which indicates that devices can not improve utility by manipulating bids. Besides, the more marginal security yield brought to the mobile blockchain network, the device has a higher probability of winning the auction. Thus, the honest bid is the dominant strategy. Theorem 2 is proved. ■

C. Computational Efficiency

Then, we prove that the proposed auction mechanism is bounded by the polynomial time.

Theorem 3: The reverse auction is Computational Efficiency and the computational complexity is $O(N^2 \log N)$.

Proof: In Algorithm 1, to determine the quality of connectivity, the process has the time complexity of $O(N)$ (lines 5–14). In Algorithm 2, the time complexity of the sorting process is $O(N \log N)$ (line 9), and the selection of non-negative marginal welfare devices needs $O(N)$ time complexity (lines 15–21). In the worst case, all devices bring marginal welfare loss in Algorithm 2 (lines 23–26), and the winner selection process has the time complexity of $O(N \log N)$. Thus, Algorithm 2 needs the time complexity of $O(N \log N)$ ultimately. In Algorithm 3, for each winner, we execute the winner device selection, which needs the time complexity of $O(N^2 \log N)$.

Therefore, the proposed reverse auction mechanism is computational efficiency and bounded by the polynomial time complexity of $O(N^2 \log N)$. Theorem 3 is proved. ■

D. Approximation Ratio

In the end, we give an analysis of the approximation ratio of our reverse auction mechanism. Inspired by papers [50] and [42], we use the liner program method to derive the approximation guarantee of the proposed mechanism. We define set $\mathcal{N}^-$ to denote devices that bring marginal welfare loss, and $\hat{C}$ is the residual expected network connectivity after selecting all non-negative marginal welfare devices. Moreover, we normalize the marginal welfare Γ_i for each device $i \in \mathcal{N}^-$ in the following

$$\Gamma_i = \frac{\Gamma_i}{\min_{i \in \mathcal{N}^-} \Gamma_i}, \quad (22)$$

where the original marginal welfare is negative, and the normalized marginal welfare is positive.

According to the (14), to maximize the social welfare, we need to minimize welfare loss while meeting the network connectivity requirement. Thus, the problem is rewritten as follows:

$$\min \sum_{i \in \mathcal{N}^-} \Gamma_i \omega_i, \quad (23)$$

$$s.t. \sum_{i \in \mathcal{N}^-} \omega_i c_i \geq \hat{C}, \quad (24)$$

$$0 \leq \omega_i \leq 1, \quad \forall i \in \mathcal{N}^-. \quad (25)$$

We transform the problem of social welfare maximization into the problem (23), and it is easily verifiable that the objective of (23) does not influence the approximation ratio. The problem (23) is a linear program problem, and we can deduce its dual problem in the following:

$$\max \xi \hat{C} - \sum_{i \in \mathcal{N}^-} \kappa_i \quad (26)$$

$$s.t. \xi c_i - \kappa_i \leq \Gamma_i, \quad \forall i \in \mathcal{N}^-, \quad (27)$$

$$\xi \geq 0, \quad (28)$$

$$\kappa_i \geq 0, \quad \forall i \in \mathcal{N}^-, \quad (29)$$

where ξ and κ_i are dual variables. We define C' is the residual expected network connectivity while device i is selected as the winner, and the welfare loss indicator η_i represents the marginal welfare loss brought by device i .

$$\eta_i = \frac{\Gamma_i}{q_i \min \{C', c_i\}}. \quad (30)$$

Assuming there are k devices in $\mathcal{N}^-$ winning the auction, we sort their welfare loss indicator in ascending order.

$$\eta_1 \leq \eta_2 \leq \dots \leq \eta_k. \quad (31)$$

Then, we define constant coefficient $\vartheta = \min_{i \in \mathcal{N}^-} q_i$ and $\sigma = \max_{i \in \mathcal{N}^-} c_i$. To resolve the approximation ratio of the proposed mechanism, we first give Lemma 1 in the following:

Lemma 1: The dual problem is resolved while dual variables ξ and κ_i satisfy the following assignments for $\forall i \in \mathcal{N}^-$:

$$\xi = \frac{\vartheta \eta_k}{2\sigma},$$

$$\kappa_i = \begin{cases} 0, & i \notin \mathbf{w}, \\ \frac{\vartheta \min\{C', c_i\}(\eta_k - \eta_i)}{2\sigma}, & i \in \mathbf{w}. \end{cases}$$

Proof: It is easily proved that $\xi, \kappa_i \geq 0$ for $\forall i \in \mathcal{N}^-$, and we verify that the assignments meet the constraint (27).

1) $i \notin \mathbf{w}$: In this case, the device i does not win the auction, and $\kappa_i = 0$. Besides, the marginal welfare loss of device i is higher than that of device k .

$$\eta_k \leq \frac{\Gamma_i}{q_i \min\{C', c_i\}}. \quad (32)$$

Thus, we derive

$$\begin{aligned} \xi c_i - \kappa_i &= \frac{\vartheta c_i \Gamma_k}{2\sigma q_i \min\{C', c_k\}} - 0 \\ &\leq \frac{\vartheta c_i \Gamma_i}{2\sigma q_i \min\{C', c_i\}} \\ &\leq \frac{\vartheta}{2q_i} \times \frac{c_i}{\sigma \min\{C', c_i\}} \times \Gamma_i \leq \Gamma_i. \end{aligned} \quad (33)$$

2) $i \in \mathbf{w}$: The device i is selected as the winner, and we deduce

$$\begin{aligned} \xi c_i - \kappa_i &= \frac{\vartheta c_i \Gamma_k}{2\sigma q_k \min\{C', c_k\}} + \frac{\vartheta \Gamma_i}{2\sigma q_i} \\ &\quad - \frac{\vartheta \Gamma_k \min\{C', c_i\}}{2\sigma q_k \min\{C', c_k\}} \\ &= \frac{\vartheta \Gamma_i}{2\sigma q_i} + \frac{\vartheta \Gamma_k (c_i - \min\{C', c_i\})}{2\sigma q_k \min\{C', c_k\}}. \end{aligned} \quad (34)$$

Case 1: $i \neq k$. Device i is not the last winner to meet the expected network connectivity, which means $c_i \leq C'$. We rewrite the (34)

$$\xi c_i - \kappa_i = \frac{\vartheta \Gamma_i}{2\sigma q_i} + \frac{\vartheta \Gamma_k (c_i - c_i)}{2\sigma q_k \min\{C', c_k\}} \quad (35)$$

$$= \frac{\vartheta \Gamma_i}{2\sigma q_i} \leq \Gamma_i. \quad (36)$$

Case 2: $i = k$. Device i is the last winner to meet the expected network connectivity, which means $c_i \geq C'$. We can derive the (34) as follows

$$\begin{aligned} \xi c_i - \kappa_i &= \frac{\vartheta \Gamma_i}{2\sigma q_i} + \frac{\vartheta \Gamma_k (c_i - C')}{2\sigma q_k \min\{C', c_k\}} \\ &\leq \frac{\Gamma_i}{2} + \frac{\Gamma_i}{2} = \Gamma_i. \end{aligned} \quad (37)$$

Therefore, the assignments are feasible for the dual problem (26) for $\forall i \in \mathcal{N}^-$, and Lemma 1 is proved. ■

Based on Lemma 1, we give the approximation ratio of the proposed mechanism in Theorem 4.

Theorem 4: The proposed reverse auction mechanism is a 2σ -approximation algorithm for the social welfare maximization problem.

Proof: Substituting the dual solutions of Lemma 1 into the (26), we can deduce

$$\begin{aligned} \xi \hat{C} - \sum_{i \in \mathcal{N}^-} \kappa_i &= \frac{\vartheta \hat{C} \Gamma_k}{2\sigma q_k \min\{C', c_k\}} - \sum_{i \in \mathcal{N}^-} \frac{\vartheta \min\{C', c_i\}}{2\sigma} \\ &\quad \times \left(\frac{\Gamma_k}{q_k \min\{C', c_k\}} - \frac{\Gamma_i}{q_i \min\{C', c_i\}} \right) \\ &= \sum_{i \in \mathcal{N}^-} \frac{\vartheta \Gamma_i}{2\sigma q_i} \leq \frac{1}{2\sigma} \sum_{i \in \mathcal{N}^-} \Gamma_i. \end{aligned} \quad (38)$$

Based on the characteristic between dual problem and primal problem, we have

$$\frac{1}{2\sigma} \sum_{i \in \mathcal{N}^-} \Gamma_i \leq \mathbf{D}^* \leq \mathbf{P}^* \leq \mathbf{S}^*, \quad (39)$$

where $\mathbf{D}^*$, $\mathbf{P}^*$, and $\mathbf{S}^*$ are optimal solutions for dual problem (26), primal problem (23), and original social welfare maximization (14), respectively. The approximation ratio Ar is derived as follows:

$$Ar = \max \left\{ \frac{\sum_{i \in \mathcal{N}^-} \Gamma_i}{\mathbf{S}^*}, \frac{\mathbf{S}^*}{\sum_{i \in \mathcal{N}^-} \Gamma_i} \right\} = 2\sigma. \quad (40)$$

Thus, the reverse auction algorithm has the approximation ratio 2σ , and Theorem 4 is proved. ■

VI. SIMULATION RESULTS

In this section, we conduct simulations to illustrate the mobile blockchain performance improvements under the proposed reverse auction mechanism. Besides, we validate the IR, IC, and the approximation ratio of the reverse auction. We use the Simpy library¹ to construct the mobile blockchain system and set up the network size as 100, where 80% are light nodes and 20% are full nodes. The upload and download bandwidth are both 2 MHz/s. The channel fading factor α follows the Rayleigh distribution with unit mean. We set up the free space path loss according to [51], where the antenna gain A_d is 4.11, the carrier frequency f_c is 915MHz, and the path loss exponent d_e is 2.8. Each device's transmission power p follows the Normal distribution with 0.5 mean and 0.1 variance, and the receiver noise power n_0 is 10^{-12} . The maximum transmission power is set to 1 Watts. The initial horizontal and vertical coordinates of mobile devices are randomly distributed in the range of $(-100, 100)$ meters. Table II lists the mobile blockchain parameters. All the simulations are performed on the desktop with an Intel Core i5-11300H 3.1 GHz CPU and 16GB memory.

A. Mobile Blockchain Performance Under Reverse Auction

First, we verify the stale rate estimation function (9). We conduct experiments on the relationship between network connectivity and the stale rate of mobile blockchain without device

¹SimPy: <https://github.com/cristiklein/simpy>

TABLE II
DEFAULT PARAMETER VALUES

Parameters	Values
Number of full nodes	20
Number of light nodes	80
Synchronization interval	30s
Block interval	3s
TTL	10
Fanout	3
Number of confirmation blocks	6
Block size	1MB
Reverse auction interval	150s
Utility coefficient γ	10
Energy cost coefficient ε	1

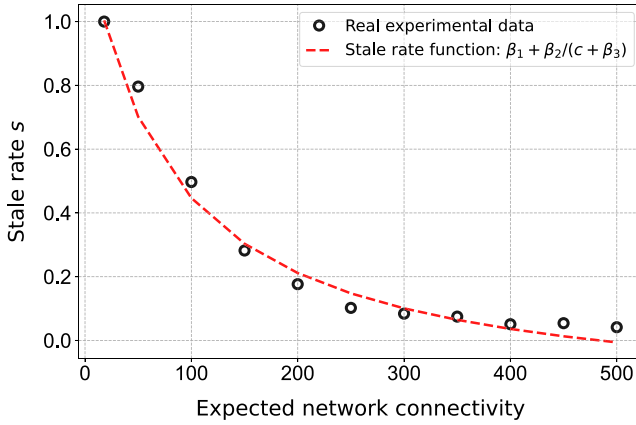


Fig. 2. Estimation of stale rate in equation (6).

movement. According to graph theory, the minimum connectivity graph of K nodes has $K - 1$ wireless connections at least. In the mobile blockchain, the only means of eliminating the stale rate is to guarantee that all full nodes are connected, where the network has at least 19 wireless connections. Thus, the initial connectivity is set to 18, which corresponds to the stale rate of 1. The R-squared of the proposed estimation function is 0.9823, where $\beta_1 = -0.2078$, $\beta_2 = 116.7171$, and $\beta_3 = 78.3394$. From the Fig. 2, we observe that the stale rate decreases with increasing network connectivity. When the number of wireless connections is greater than 300, the decline of the stale rate becomes insignificant.

We compare the proposed mechanism with the following schemes:

- Scheme 1: selfless static device. Each device does not move and is willing to help propagate blocks, and the wireless connections of the blockchain network are stable.
- Scheme 2: selfless moving device. Each device moves with the 1 m/s velocity and is willing to help propagate blocks. However, the wireless connections of the blockchain network are unstable and change due to devices' movement.
- Scheme 3: selfish moving device. Each device moves with the 1 m/s velocity and only propagates its own blocks. The wireless connections of the blockchain network also change with devices' movement.

In the proposed mechanism, each device moves at 1 m/s. The winners of the reverse auction consume their energy to disseminate blocks and get payments accordingly.

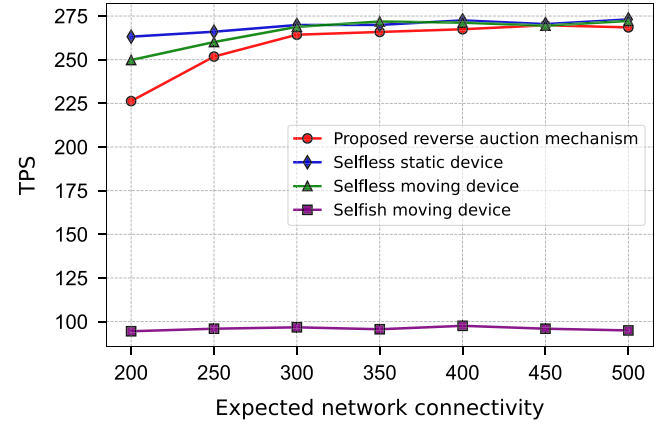


Fig. 3. TPS under different schemes.

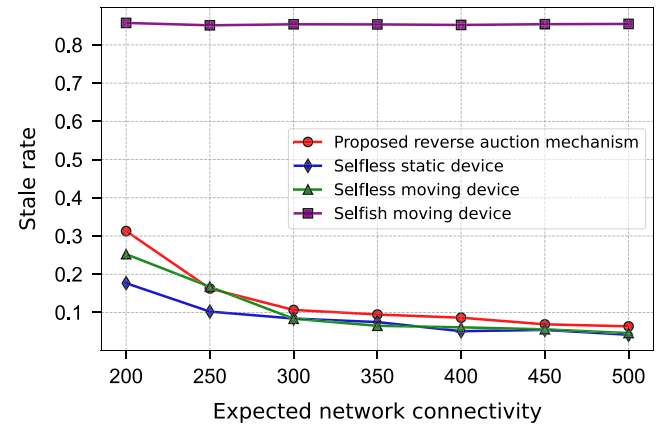


Fig. 4. Stale rate under different schemes.

Scheme 1 represents the optimal performance of the blockchain network without device movement. Conversely, scheme 2 demonstrates the best performance of blockchain in a dynamic mobile network setting.

Fig. 3 demonstrates the Throughput per Second (TPS) of the mobile blockchain system under four schemes. The connectivity demand C is the total wireless connections of the blockchain network. The TPS of the mobile blockchain increases with the connectivity demand under selfless schemes and the proposed mechanism. In the selfish scheme, mobile devices just care about their own blocks. Even with increased connectivity, they remain reluctant to help propagate blocks due to the associated energy consumption. Moreover, the blockchain's competitive block generation mechanism and network propagation latency result in an upper limit on the throughput of the blockchain network.

In Fig. 4, we give the variation of the stale rate with network connectivity. Intuitively, the selfish scheme gets the worst result, and the stale rate of the mobile blockchain network is as high as 85%, which means the blockchain is insecure and can't operate properly. The proposed mechanism effectively incentivizes mobile devices to contribute their wireless connections to the blockchain network, which achieves a stale rate similar to schemes 1 and 2. Similarly, the stale rate cannot decrease infinitely with increasing wireless connections.

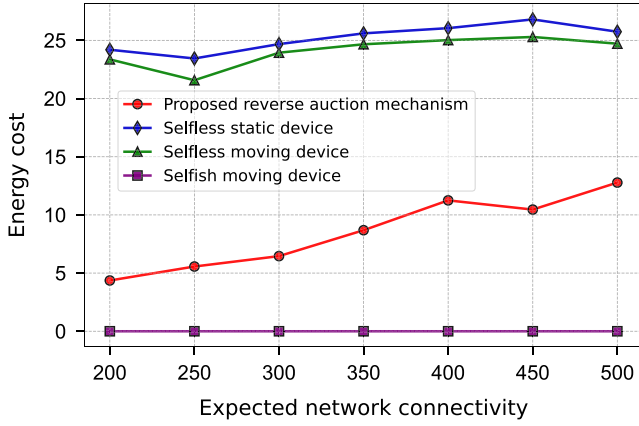


Fig. 5. Energy cost under different schemes.

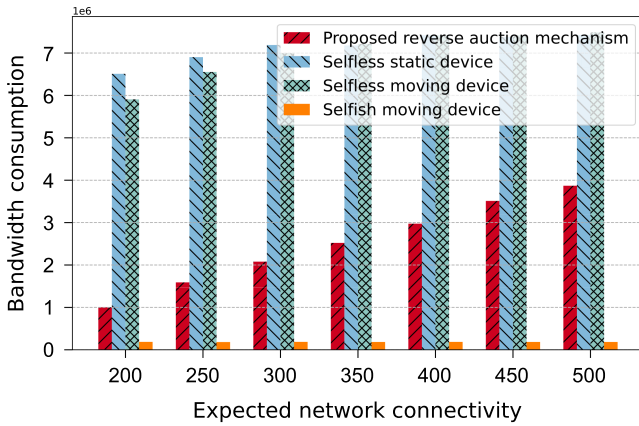


Fig. 6. Bandwidth consumption under different schemes.

Fig. 5 shows the energy cost under different schemes. We define energy cost as the power consumption of the wireless device when it helps transfer others' blocks. Since the device just transfers its own blocks in the selfish scheme, the energy cost of the mobile blockchain network is 0. The two selfless schemes have similar energy costs, with the no-movement scheme having a slightly higher energy cost. The devices in the selfless and mobile scheme are in the dynamic mobile network, which leads to disconnections, making the total number of wireless connections in the mobile blockchain network less than the expected value thus slightly reducing the energy cost. Our proposed mechanism requires only the winners of the reverse auction to help in the information dissemination of the mobile blockchain network, which drastically reduces the energy cost.

In Fig. 6, we compare the bandwidth consumption for propagating blocks over different schemes. The selfish scheme has the lowest bandwidth consumption, which sacrifices the performance and security of the blockchain system accordingly. When the network connectivity is no more than 300, the bandwidth consumption of the proposed mechanism is no greater than $\frac{1}{3}$ compared to selfless schemes. While the network connectivity is greater than 300, the proposed mechanism consumes only half of the bandwidth of selfless schemes.

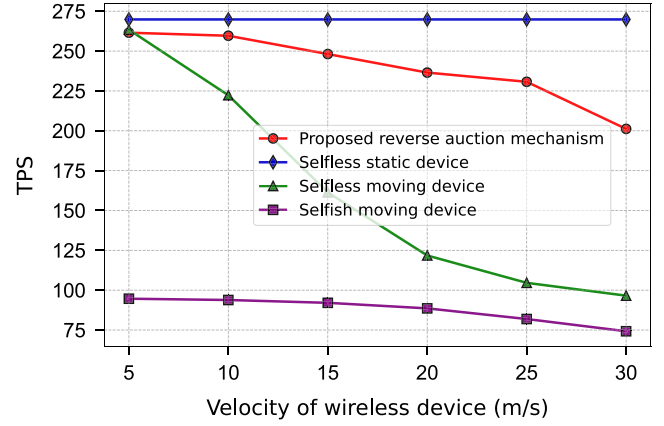


Fig. 7. Impact of device velocity on TPS.

From Fig. 3–6, comparing with selfless schemes, the proposed reverse auction mechanism achieves similar TPS and stale rates but with up to half the energy and bandwidth consumption in the dynamic mobile blockchain network.

B. Impact of Device Velocity on Mobile Blockchain Performance

In this subsection, we discuss the impact of device mobility on blockchain performance. We set the velocity of the device from 5 m/s to 30 m/s and observe the performance variation of the mobile blockchain under different mechanisms. Devices don't move in the selfless and no-movement scheme, and we set it as the optimal TPS and stale rate performances that a mobile blockchain system can achieve.

In Fig. 7, we observe that the TPS of the mobile blockchain network decreases as the node speed increases. Since wireless connection interruption is frequent with high device mobility, the information of mobile blockchain consumes more time to propagate and achieve consensus, which decreases the TPS. Though mobile devices help relay blockchain information in the selfless scheme, high-velocity movement makes devices unable to maintain stable connections, resulting in sharp TPS degradation. In our proposed mechanism, we use quality of connectivity to select devices with robust wireless connections and give rewards to incentivize devices to rebuild connections while link interruption occurs. Thus, the proposed mechanism can reduce the impact of dynamic networks on the TPS of mobile blockchain.

Fig. 8 shows the variation of stale rate with devices' velocity. The higher velocity of devices brings a higher stale rate of mobile blockchain. Our designed mechanism greatly reduces the impact of device velocity on the stale rate. When the device velocity is less than 20 m/s, the proposed maintains the stale rate of mobile blockchain below 30%, which ensures the security of the mobile blockchain system. Even if mobile devices are in hypervelocity (beyond 20 m/s), our mechanism still ensures that the stale rate of mobile blockchain is under 50%. In the scheme of the selfless device with movement, the stale rate of blockchain is more than 69% under device velocity of 15 m/s and over 80%

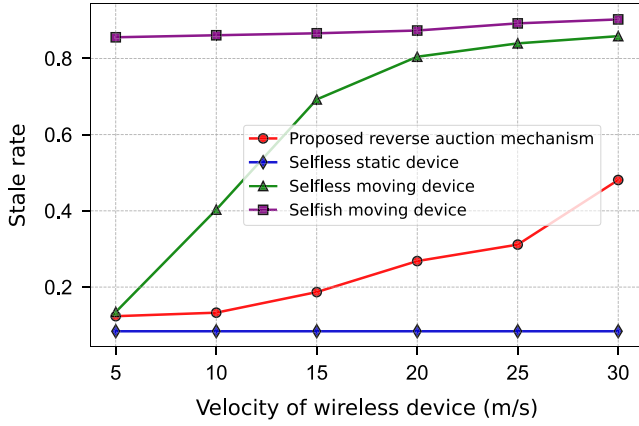


Fig. 8. Impact of device velocity on stale rate.

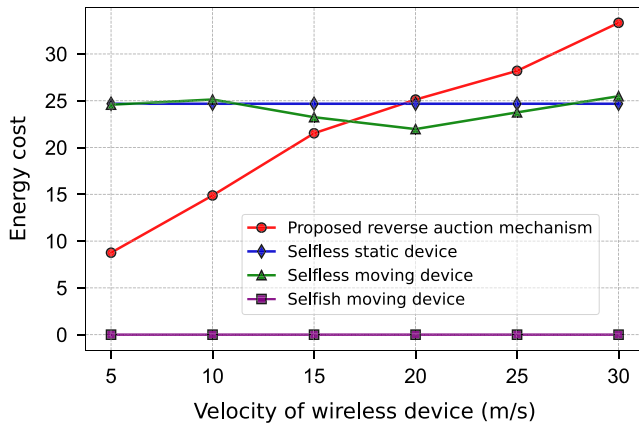


Fig. 9. Impact of device velocity on energy cost.

while device velocity is beyond 20 m/s, which means the existing blockchain mechanism does not apply to dynamic mobile networks. Thus, the proposed reverse auction mechanism makes up for the shortcomings of the existing blockchain mechanism that cannot guarantee stable connections in high mobile networks and reduces the stale rate of mobile blockchain.

Fig. 9 demonstrates the impact of device velocity on the energy cost of the mobile blockchain system. In the reverse auction mechanism, as the velocity increases, mobile devices' retransmission increases, resulting in higher energy costs. In the selfless scheme, on the one hand, device mobility reduces the amount of wireless connections and devices receive fewer messages to relay. On the other hand, link interruption increases the number of retransmissions. Thus, the energy cost of mobile blockchain fluctuates with the velocity under the scheme of the selfless device with movement.

In Fig. 10, we gave the variation of bandwidth consumption under different device velocities. The proposed mechanism selects devices with robust wireless connections to maintain stable network connectivity, where the bandwidth consumption is almost unfluctuating as velocity changes. In scheme 2, the mobility of the device reduces the wireless connection amount and decreases the bandwidth consumption accordingly.

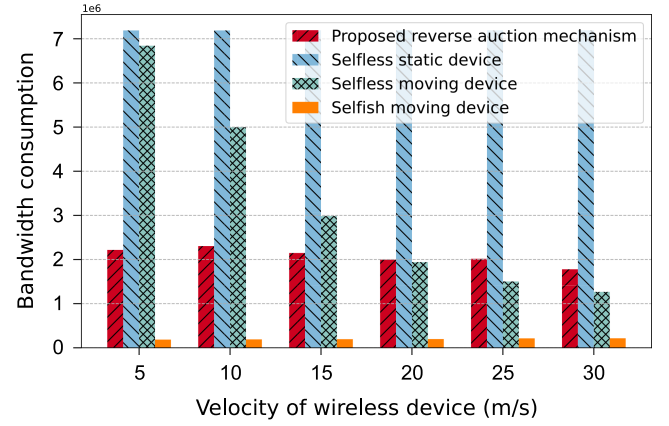


Fig. 10. Impact of device velocity on bandwidth consumption.

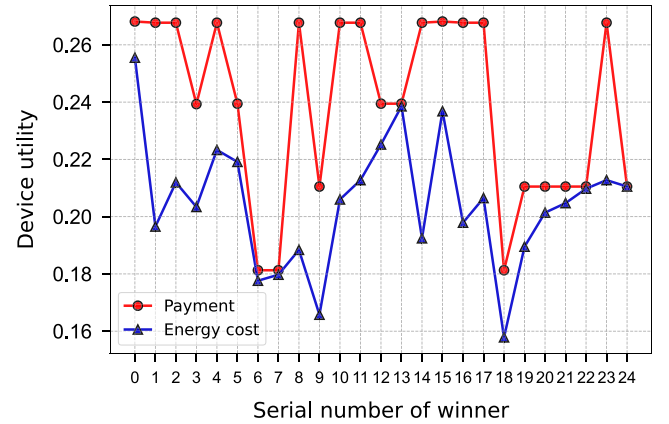


Fig. 11. IR validation of the proposed mechanism.

C. Validation of Reverse Auction

In this subsection, we conduct simulations to verify the IR, IC, CE, and approximation ratio of the proposed reverse auction.

Fig. 11 shows the IR validation of the proposed mechanism. We start the reverse auction and record the payment and energy cost of the winning devices. From Fig. 11, we observe that each winner obtains the payment with no less than its energy cost, which means the mobile device gets non-negative utility by participating in the reverse auction.

In Fig. 12, we select two devices to validate the IC of the reverse auction, where device 3 is the winner of the auction and device 7 loses the auction. The bid of the device includes the wireless connections offered and the reward value it expects to receive. We define the true reward value as the energy cost of the device, and the utility of the device is calculated according to (12). We alter the reward value of the bid to observe the change in device utility. From the perspective of device 3, despite the reduction in the reward value, device 3 remains the winner. However, this does not increase the utility. If device 3 increases the reward value, it will lose the auction and get zero utility. As for device 7, its true bid does not win the auction, and increasing the reward value still results in the auction being lost with zero utility. If device 7 reduces the reward value, it will win the auction. However, the payment for device 7 does not compensate

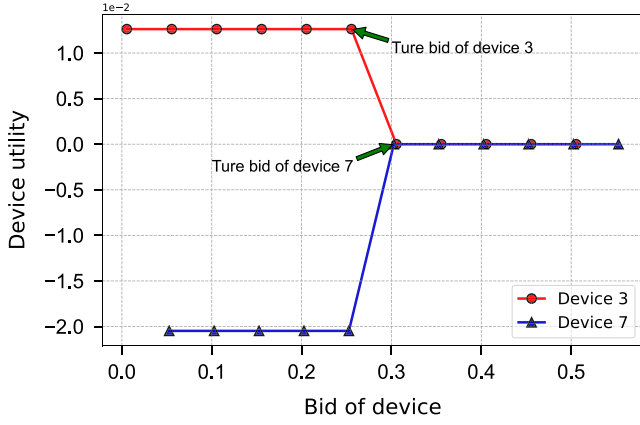


Fig. 12. IC validation of the proposed mechanism.

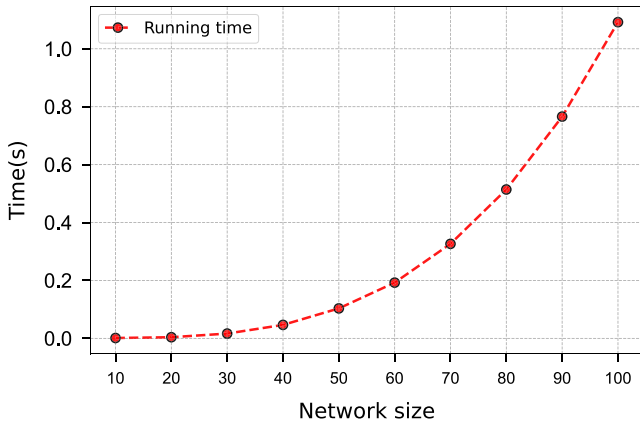


Fig. 13. Computational efficiency validation of the proposed mechanism.

for the energy cost, and device 7 obtains negative utility. Thus, the dominant strategy is honest bidding, which maximizes the utility of mobile devices.

Fig. 13 demonstrates the variation of the running time of the proposed mechanism with the network size. We observe that the running time changes exponentially with the increase in network size, which is consistent with the theoretical result $O(N^2 \log N)$. Thus, our reverse auction is computational efficiency.

In Section V, we prove that the reverse auction is the 2σ -approximation algorithm for optimal social welfare. In Fig. 14, we set the maximal wireless connection in the bidding from 1 to 10, take the social welfare obtained by the exhaustive indexing method as the optimal social welfare, and define the actual approximation ratio as the ratio of the optimal social welfare to the social welfare obtained by the proposed mechanism. From Fig. 14, the actual approximation ratio is always lower than the theoretical value validating the correctness of the proof. Besides, the maximum actual approximation ratio is 1.09, which means the social welfare of the proposed mechanism is very close to the optimal value.

Fig. 15 compares the social welfare among the proposed mechanism, the exhaustive indexing method, and the Bitcoin mechanism under different connection demands. We observe that the reverse auction achieves optimal social welfare under 100, 110, 120, 130, 170, and 200 network connectivity. Even

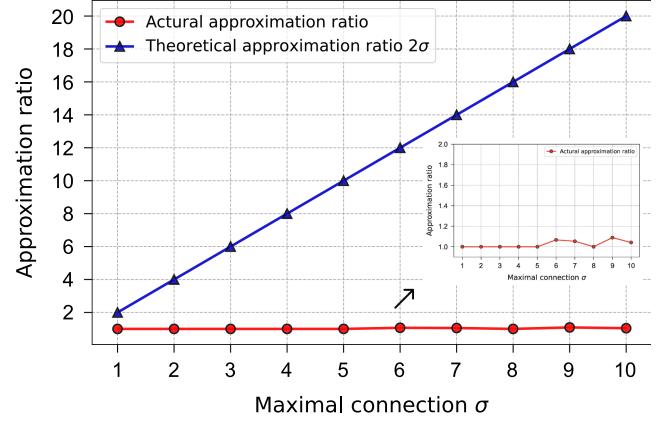


Fig. 14. The approximation ratio validation of the proposed mechanism.

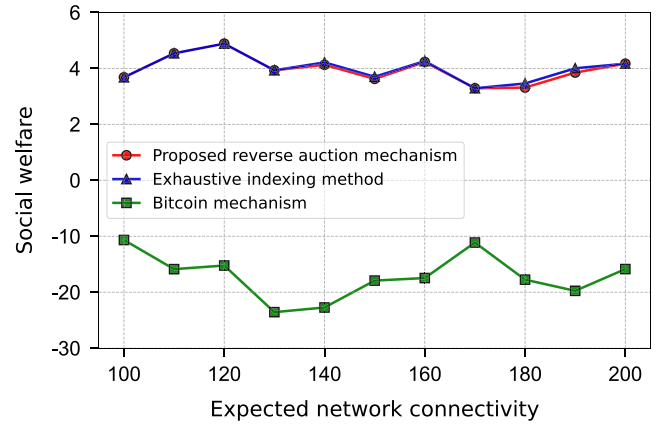


Fig. 15. The social welfare of the blockchain system under different mechanisms.

in the worst case (connectivity 190), the proposed mechanism achieves 96.19% of the optimal social welfare. The Bitcoin mechanism gets the worst social welfare in the mobile blockchain network. Since the Bitcoin mechanism is unable to optimize wireless connections of devices based on network dynamics, its energy efficiency is low, resulting in negative social welfare.

D. Impact of Block Size and Interval on Mobile Blockchain Performance

Fig. 16 demonstrates how the block size and interval affect the mobile blockchain performance. We set the expected network connectivity in the mobile blockchain to 300 and the device velocity to 5 m/s. Fig. 16(a) illustrates the trend of TPS as block interval and block size vary. From Fig. 16(a), we observe that block size has a more significant impact on TPS. As block size increases, more transactions can be included in each block, enhancing the TPS of mobile blockchain. Conversely, longer block intervals result in fewer blocks being generated per unit of time, thereby reducing throughput.

Fig. 16(b) demonstrates the impact of block interval and block size on block stale rate. As block interval increases, devices have more time for consensus and block propagation, reducing forking events and consequently lowering the block stale rate.

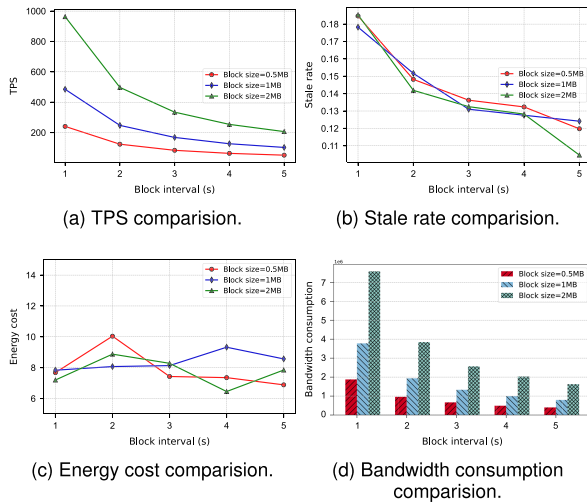


Fig. 16. The performance of mobile blockchain under different block intervals and block sizes.

However, block size shows no significant effect on the block stalen rate.

As shown in Fig. 16(c), the energy cost in the network is not affected by block interval or block size, since we define energy cost as the energy consumed by devices to maintain wireless connections. Therefore, the energy cost primarily depends on the velocity of devices and the number of wireless connections.

In Fig. 16(d), we observe that bandwidth consumption in the mobile blockchain increases with block size and decreases with block interval. Similar to Fig. 16(a), larger block sizes require more bandwidth resources for block propagation, resulting in higher bandwidth consumption. Meanwhile, longer block interval reduce the frequency of block propagation, thereby decreasing bandwidth consumption.

VII. CONCLUSION

In this paper, we investigate how to select devices to ensure stable performance of the blockchain system under the dynamic mobile network with fast-moving devices. We design a reverse auction mechanism to autonomously select devices with robust wireless connections to maintain the blockchain network, where wireless devices submit their wireless connections and expected rewards as bids. By elaborating the pricing algorithm, the proposed mechanism ensures IR, IC, and CE. Besides, the proposed mechanism achieves more than 96% of optimal social welfare. In future work, we will explore the impact of more complex wireless environments on mobile blockchain performance and design appropriate mechanisms to ensure the efficient operation of the mobile blockchain system.

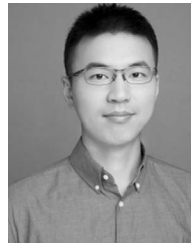
REFERENCES

- [1] T. Hewa, G. Gür, A. Kalla, M. Ylianttila, A. Bracken, and M. Liyanage, "The role of blockchain in 6G: Challenges, opportunities and research directions," in *Proc. 2nd 6G Wireless Summit*, 2020, pp. 1–5.
- [2] Y. Zeng et al., "A tutorial on environment-aware communications via channel knowledge map for 6G," *IEEE Commun. Surveys Tuts.*, vol. 26, no. 3, pp. 1478–1519, Third Quarter 2024.
- [3] A. Hammoud, H. Sami, A. Mourad, H. Otrok, R. Mizouni, and J. Bentahar, "AI, blockchain, and vehicular edge computing for smart and secure IoV: Challenges and directions," *IEEE Internet Things Mag.*, vol. 3, no. 2, pp. 68–73, Jun. 2020.
- [4] F. Liu, R. Sun, D. Wang, C. Javali, and P. Liu, "6G native trust," 2022. [Online]. Available: <https://www-file.huawei.com/-/media/corp2020/pdf/publications/huawei-research/issue2/6g-native-trustworthiness-cn.pdf?la=zh>
- [5] I. Al Ridhawi, M. Aloqaily, and F. Karray, "Intelligent blockchain-enabled communication and services: Solutions for moving Internet of Things Devices," *IEEE Robot. Automat. Mag.*, vol. 29, no. 2, pp. 10–20, Jun. 2022.
- [6] G. Karagiannis et al., "Vehicular networking: A survey and tutorial on requirements, architectures, challenges, standards and solutions," *IEEE Commun. Surveys Tuts.*, vol. 13, no. 4, pp. 584–616, Fourth Quarter 2011.
- [7] G. Jing, Y. Zou, Z. Zhang, D. Yu, F. Dressler, and X. Cheng, "Byzantine fault tolerant consensus in open wireless networks via an abstract MAC layer," *IEEE Trans. Commun.*, vol. 73, no. 3, pp. 1909–1924, Mar. 2025.
- [8] S. Nakamoto, "Bitcoin: A peer-to-peer electronic cash system," 2008. [Online]. Available: <http://www.bitcoin.org/bitcoin.pdf>
- [9] V. Buterin et al., "Ethereum: A next generation smart contract and decentralized application platform," 2013. [Online]. Available: <http://ethereum.org/ethereum.html>
- [10] Q. Bai, X. Zhou, X. Wang, Y. Xu, X. Wang, and Q. Kong, "A deep dive into blockchain selfish mining," in *Proc. IEEE Int. Conf. Commun.*, 2019, pp. 1–6.
- [11] W. N. Suliyanti and R. F. Sari, "Evaluation of hash rate-based double-spending based on proof-of-work blockchain," in *Proc. Int. Conf. Inf. Commun. Technol. Convergence*, 2019, pp. 169–174.
- [12] S. D. Jap, "Online reverse auctions: Issues, themes, and prospects for the future," *J. Acad. Marketing Sci.*, vol. 30, pp. 506–525, 2002.
- [13] J. Wang, X. Ling, Y. Le, Y. Huang, and X. You, "Blockchain-enabled wireless communications: A new paradigm towards 6G," *Nat. Sci. Rev.*, vol. 8, no. 9, 2021, Art. no. nwab069.
- [14] Z. Su, Y. Wang, Q. Xu, and N. Zhang, "LVBS: Lightweight vehicular blockchain for secure data sharing in disaster rescue," *IEEE Trans. Dependable Secure Comput.*, vol. 19, no. 1, pp. 19–32, Jan./Feb. 2022.
- [15] T. Wu et al., "Blockchain-based anonymous data sharing with accountability for Internet of Things," *IEEE Internet Things J.*, vol. 10, no. 6, pp. 5461–5475, Mar. 2023.
- [16] W. Lu, Z. Ren, J. Xu, and S. Chen, "Edge blockchain assisted lightweight privacy-preserving data aggregation for smart grid," *IEEE Trans. Netw. Service Manag.*, vol. 18, no. 2, pp. 1246–1259, Jun. 2021.
- [17] S. Aggarwal, N. Kumar, and S. Tanwar, "Blockchain-envisioned UAV communication using 6G networks: Open issues, use cases, and future directions," *IEEE Internet Things J.*, vol. 8, no. 7, pp. 5416–5441, Apr. 2021.
- [18] J. Shi, J. Du, Y. Shen, J. Wang, J. Yuan, and Z. Han, "DRL-based V2V computation offloading for blockchain-enabled vehicular networks," *IEEE Trans. Mobile Comput.*, vol. 22, no. 7, pp. 3882–3897, Jul. 2023.
- [19] X. Tang, Q. Chen, R. Yu, and X. Li, "Digital twin-empowered task assignment in aerial MEC network: A resource coalition cooperation approach with generative model," *IEEE Trans. Netw. Sci. Eng.*, vol. 12, no. 1, pp. 13–27, Jan./Feb. 2025.
- [20] C. Xu, P. Zhang, X. Xia, L. Kong, P. Zeng, and H. Yu, "Digital twin-assisted intelligent secure task offloading and caching in blockchain-based vehicular edge computing networks," *IEEE Internet Things J.*, vol. 12, no. 4, pp. 4128–4143, Feb. 2025.
- [21] J. Huang et al., "Blocksense: Towards trustworthy mobile crowdsensing via proof-of-data blockchain," *IEEE Trans. Mobile Comput.*, vol. 23, no. 2, pp. 1016–1033, Feb. 2024.
- [22] N. C. Luong, T. Le Van, S. Feng, H. Du, D. Niyato, and D. I. Kim, "Edge computing for metaverse: Incentive mechanism versus semantic communication," *IEEE Trans. Mobile Comput.*, vol. 23, no. 5, pp. 6196–6211, May 2024.
- [23] N. C. Luong et al., "Optimal auction for effective energy management in UAV-assisted vehicular metaverse synchronization systems," *IEEE Trans. Veh. Technol.*, vol. 73, no. 1, pp. 1207–1222, Jan. 2024.
- [24] J. Huang et al., "Secure data sharing over vehicular networks based on multi-sharding blockchain," *ACM Trans. Sen. Netw.*, vol. 20, no. 2, pp. 1–23, Jan. 2024.
- [25] J. Huang, L. Kong, G. Chen, Q. Xiang, X. Chen, and X. Liu, "Blockchain-based federated learning: A systematic survey," *IEEE Netw.*, vol. 37, no. 6, pp. 150–157, Nov. 2023.

- [26] Y. Xiao, N. Zhang, W. Lou, and Y. T. Hou, "Modeling the impact of network connectivity on consensus security of proof-of-work blockchain," in *Proc. IEEE Conf. Comput. Commun.*, 2020, pp. 1648–1657.
- [27] J. Mišić, V. B. Mišić, X. Chang, S. G. Motlagh, and M. Z. Ali, "Modeling of bitcoin's blockchain delivery network," *IEEE Trans. Netw. Sci. Eng.*, vol. 7, no. 3, pp. 1368–1381, Third Quarter 2020.
- [28] P. Zhang, Y. Tao, Q. Zhao, and M. Zhou, "A rate-and-trust-based node selection model for block transmission in blockchain networks," *IEEE Internet Things J.*, vol. 10, no. 2, pp. 1605–1616, Jan. 2023.
- [29] G. Lee, J. Park, W. Saad, and M. Bennis, "Performance analysis of blockchain systems with wireless mobile miners," *IEEE Netw. Lett.*, vol. 2, no. 3, pp. 111–115, Sep. 2020.
- [30] O. Onireti, L. Zhang, and M. A. Imran, "On the viable area of wireless practical byzantine fault tolerance (PBFT) blockchain networks," in *Proc. IEEE Glob. Commun. Conf.*, 2019, pp. 1–6.
- [31] X. Hou, Y. Li, D. Jin, D. O. Wu, and S. Chen, "Modeling the impact of mobility on the connectivity of vehicular networks in large-scale urban environments," *IEEE Trans. Veh. Technol.*, vol. 65, no. 4, pp. 2753–2758, Apr. 2016.
- [32] O. Rehman, R. Qureshi, M. Ould-Khaoua, and M. F. Niazi, "Analysis of mobility speed impact on end-to-end communication performance in vanets," *Veh. Commun.*, vol. 26, 2020, Art. no. 100278.
- [33] X. Zhang et al., "The block propagation in blockchain-based vehicular networks," *IEEE Internet Things J.*, vol. 9, no. 11, pp. 8001–8011, Jun. 2022.
- [34] A. Anand, W. Asif, and M. Lestas, "Performance evaluation of pow blockchain in wireless mobile IoT networks," in *Proc. 17th Int. Conf. Distrib. Comput. Sensor Syst.*, 2021, pp. 396–403.
- [35] S. Kim, "Impacts of mobility on performance of blockchain in vanet," *IEEE Access*, vol. 7, pp. 68 646–68 655, 2019.
- [36] R. Han, Z. Yan, X. Liang, and L. T. Yang, "How can incentive mechanisms and blockchain benefit with each other? A survey," *ACM Comput. Surv.*, vol. 55, no. 7, pp. 1–38, 2022.
- [37] V. Buterin and V. Griffith, "Casper the friendly finality gadget," 2017, *arXiv: 1710.09437*.
- [38] V. Buterin, D. Reijnders, S. Leonardos, and G. Piliouras, "Incentives in ethereum's hybrid casper protocol," in *Proc. IEEE Int. Conf. Blockchain Cryptocurrency*, 2019, pp. 236–244.
- [39] V. Buterin et al., "Combining ghost and casper," 2020, *arXiv: 2003.03052*.
- [40] W. Wang, J. Chen, Y. Jiao, J. Kang, W. Dai, and Y. Xu, "Connectivity-aware contract for incentivizing IoT devices in complex wireless blockchain," *IEEE Internet Things J.*, vol. 10, no. 12, pp. 10 413–10 425, Jun. 2023.
- [41] J. Li, T. Liu, D. Niyato, P. Wang, J. Li, and Z. Han, "Contract-theoretic pricing for security deposits in sharded blockchain with Internet of Things (IoT)," *IEEE Internet Things J.*, vol. 8, no. 12, pp. 10 052–10 070, Jun. 2021.
- [42] H. Jin, L. Su, D. Chen, K. Nahrstedt, and J. Xu, "Quality of information aware incentive mechanisms for mobile crowd sensing systems," in *Proc. 16th ACM Int. Symp. Mobile Ad Hoc Netw. Comput.*, 2015, pp. 167–176.
- [43] Y. Jiao, P. Wang, D. Niyato, and K. Suankawmanee, "Auction mechanisms in cloud/fog computing resource allocation for public blockchain networks," *IEEE Trans. Parallel Distrib. Syst.*, vol. 30, no. 9, pp. 1975–1989, Sep. 2019.
- [44] W. Wang, Y. Jiao, J. Chen, W. Dai, J. Kang, and Y. Xu, "Dual auction mechanism for transaction forwarding and validation in complex wireless blockchain network," 2023, *arXiv:2309.01920*.
- [45] T. S. Rappaport, *Wireless Communications: Principles and Practice*. Cambridge, U.K.: Cambridge Univ. Press, 2024.
- [46] C. Decker and R. Wattenhofer, "Information propagation in the bitcoin network," in *Proc. IEEE Int. Conf. Peer-to-Peer Comput.*, 2013, pp. 1–10.
- [47] A. Miller, Y. Xia, K. Croman, E. Shi, and D. Song, "The honey badger of BFT protocols," in *Proc. ACM SIGSAC Conf. Comput. Commun. Secur.*, New York, NY, USA, 2016, pp. 31–42.
- [48] Y. Jiao, P. Wang, D. Niyato, B. Lin, and D. I. Kim, "Toward an automated auction framework for wireless federated learning services market," *IEEE Trans. Mobile Comput.*, vol. 20, no. 10, pp. 3034–3048, Oct. 2021.
- [49] A. Gervais, G. O. Karame, K. Wüst, V. Glykantzis, H. Ritzdorf, and S. Capkun, "On the security and performance of proof of work blockchains," in *Proc. ACM SIGSAC Conf. Comput. Commun. Secur.*, 2016, pp. 3–16.
- [50] S. Rajagopalan and V. Vazirani, "Primal-dual rnc approximation algorithms for (multi)-set (multi)-cover and covering integer programs," in *Proc. IEEE 34th Annu. Found. Comput. Sci.*, 1993, pp. 322–331.
- [51] L. Huang, S. Bi, and Y.-J. A. Zhang, "Deep reinforcement learning for on-line computation offloading in wireless powered mobile-edge computing networks," *IEEE Trans. Mobile Comput.*, vol. 19, no. 11, pp. 2581–2593, Nov. 2020.



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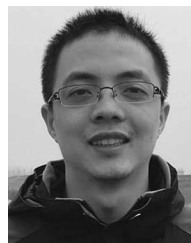
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